

C2_03 Gate C Protected Review - Astra RC4

Logistics Shipments & Service Targets. Final fresh reassessment; open only after saved Gate C evidence.

1. Objective controls

Control	Expected	Why it matters
Fact rows	Shipments 100 Events 233	Events are one-to-many under shipments: 2-3 events per shipment.
Hub key	8 raw / 7 unique duplicate H02	One-side hub must be unique before relationship.
Unknown hub	H99 on shipment S096 charge 594	Dropping it changes charge total 102,950 -> 102,356.
Event distribution	100 Booked + 100 Delivered + 33 DelayFlag	Explains 233 event rows without changing shipment count.
Fan-out trap	Shipment charge 102,950; after naive event join/sum = 239,400	Concrete proof that event grain can multiply shipment-level measures.
Date-role contrast	Book rows Mar/Apr/May/Jun = 18/30/30/22; Delivery = 15/29/31/25	Booking and delivery period questions differ.
Targets	21 Hub-Month rows target units 3360	Seven hubs x three months at higher grain; do not repeat across shipments/events.

2. Expected reasoning path

FactShipment is one shipment; FactShipmentEvent is one event. The event fact contains 233 rows for 100 shipments. A direct join followed by SUM(Shipment[Charge]) would inflate 102,950 to 239,400 - a model-grain failure, not a DAX formatting issue.

H02 is an exact duplicate in the dirty hub file; prove identity before deterministic dedupe. H99 on delivered shipment S096 must remain visible because silently losing it removes 594 of charge.

Booked and Delivered date roles shift monthly populations. Default/alternate date behavior must be tied to the report question and proven with known boundary shipments.

Hub-Month targets are higher-grain facts. Model them through conformed Hub and Date/Month dimensions; never attach them to shipment events.

Event tests should show two baseline events per shipment plus a DelayFlag only for delayed shipments. Shipment status control is 67 Delivered / 33 Delayed; event counts and shipment status must stay conceptually separate.

3. Critical failures

- Sums shipment-level charge at event grain.
- Uses Both/M:M to make H02 “work” without proving the key repair.
- Drops H99 and accepts 102,356 as the model total.
- Uses EventDate as a substitute for both BookDate and DeliveryDate without a role contract.
- Repeats Hub-Month target across shipments or events.
- Optimizes or publishes before the model reconciles to source rows/keys/totals.

4. Competency floors

Competency	Required proof
Model contract / grain	Every fact and higher-grain table has an explicit row meaning; no mixed-grain shortcut.
Keys / star design	One-side uniqueness and orphan handling are evidenced before relationships; no arbitrary dedupe.
Relationships / date roles	Cardinality, active state and direction documented; tiny filter tests prove both required date roles.
Power Query / reconciliation	Traceable source-staging-model-ready path plus independent rows/keys/totals/exceptions.
Performance / handoff	Correctness first; runtime performance evidence only when actually observed; RELEASE/BLOCK and repair path are explicit.

Changed repair proof

If instability remains after Gate C, return only to the named modeling competency with a new fixture and fresh evidence. Do not recycle Gate A/B/C answers as proof.